

lum magneticum—that eats iron to create tiny magnets inside itself. These tiny magnets are quite similar to those found in traditional hard drives.

The scientists feel that their research could lead to much faster, higher-density hard drives and a range of other high-performance, environment-friendly electronic devices.

"Nature can help us. We are forever trying to make electronic components smaller but are quickly reaching our limits with traditional manufacturing techniques," said Sara Staniland from Leeds University's School of Physics and Astronomy.

The bacterium used in the research is a natural magnetic microorganism, which is found in ponds and lakes and swims along the Earth's magnetic lines. Therefore, when it eats iron, proteins interact to produce tiny crystals of magnetite—the most magnetic of all naturally occurring minerals on Earth.

Spintronics for faster electronic devices

A team of physicists from the University of South Florida and the University of Kentucky has taken a big step toward the development of practical spintronic devices—a technology that could help create faster, smaller and more versatile electronic devices. The team showed that a simple combination of metal atoms and a flat sheet of one atom-thick layer of pure carbon called graphene can be suitably engineered and used for this purpose. The biggest challenge for the researchers was finding a semiconductor material that has a net 'spin' at room temperature and determining how to set the spin.

Using state-of-the-art theoretical computations, the research team demonstrated that by placing cobalt atoms in graphene holes, it is possible to set the spins in a controlled manner. The holes were created by removing one or two nearby carbon atoms. Setting the spin is the key to practical spintronics application for graphene.

Speckle-free lasers could power high-definition imaging

A team of researchers led by Brendan Redding of Yale University has demonstrated low-coherence 'random lasers' that produce images as speckle-free as those made using LEDs or bulbs. In these lasers, light is reflected off small particles randomly distributed in a fluid inside the laser—instead of mirrors at either end of the laser, as is typically done to concentrate the beam.

Lasers are used to power high-definition projectors in cinemas and to illuminate objects for imaging. But speckles that randomly surround the laser beam blur images. To compensate for this, complex laser-based optical systems have been devised with parts constantly in motion to suppress the mottled light. Breaking down coherence inside the laser itself is a much simpler solution.

According to team member Hui Cao, low-coherence random lasers should light up small objects well enough to produce images that track real-time motion, like the beating of the heart in an embryo.

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